

VENTERSDORP, South Africa

WMO: 683490

Lat: 26.312S

Lon: 26.814E

Elev: 4902

StdP: 12.27

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	32.7	35.4	10.4	11.2	57.7	14.2	13.5	55.9	41.1	62.7	35.3	61.8	6.8	20	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	89.7	60.9	87.4	60.4	85.1	60.5	67.5	78.0	66.7	77.0	65.8	76.0	11.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	110.5	69.7	64.1	107.7	69.4	62.9	103.2	68.9	35.1	77.9	34.3	76.8	33.6	76.4	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.1	29.9	24.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	71.5	71.0	67.9	63.1	57.1	52.6	52.4	57.0	64.1	68.0	68.7	71.2
	DBStd	8.52	4.63	3.88	3.77	4.70	5.04	4.23	4.73	5.60	6.25	6.04	5.70	4.55
	HDD50	77	0	0	0	1	12	19	30	12	2	1	0	0
	HDD65	1534	7	5	15	87	245	372	391	251	87	40	27	7
	CDD50	5068	665	589	555	393	233	97	103	230	426	559	562	656
	CDD65	1051	207	173	105	30	1	0	0	4	61	134	139	197
	CDH74	8490	1396	1141	707	246	57	2	2	113	735	1294	1316	1481
	CDH80	2387	426	329	122	30	2	0	0	6	183	414	414	461
(14) Wind	WSAvg	10.2	10.4	10.7	7.6	8.2	8.4	8.9	9.3	11.7	11.9	12.1	12.0	11.4
(15) Precipitation	PrecAvg	23.7	4.0	3.1	3.2	1.7	0.8	0.2	0.0	0.2	0.6	2.2	2.6	4.0
	PrecMax	34.4	7.3	7.4	8.4	5.0	3.9	1.6	0.3	0.9	3.0	7.5	7.2	6.1
	PrecMin	15.4	1.1	0.9	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.3	1.4
	PrecStd	4.8	1.8	1.7	1.7	1.2	1.0	0.4	0.1	0.3	0.8	1.7	1.5	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	90.7	86.1	83.6	78.3	73.0	73.7	80.6	87.1	90.8	92.7	92.7
		MCWB	62.6	62.4	60.4	60.1	53.2	50.1	50.6	55.5	56.6	58.2	58.7	62.5
	2%	DB	87.6	87.2	83.3	79.7	75.4	69.6	70.1	77.1	84.4	87.9	88.0	88.3
		MCWB	64.0	62.4	61.2	59.2	53.7	49.5	49.6	53.3	55.8	57.8	58.6	62.3
	5%	DB	84.5	84.1	80.7	76.8	73.1	66.5	67.7	73.5	81.9	85.0	84.8	84.7
		MCWB	64.0	63.3	61.7	58.3	53.2	48.5	48.8	51.0	54.8	57.5	58.9	62.5
	10%	DB	81.3	80.9	78.2	73.9	69.9	63.9	64.6	69.9	78.4	81.7	81.8	81.8
		MCWB	64.3	63.7	61.6	57.6	51.9	47.9	47.9	49.7	53.9	57.3	59.1	62.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	68.2	66.7	64.6	57.8	53.6	52.6	58.2	60.0	63.7	66.8	68.8
		MCDB	79.1	79.5	77.2	73.8	71.8	65.7	67.9	74.9	78.8	78.2	76.5	78.7
	2%	WB	67.7	66.9	65.4	62.8	56.2	51.9	51.1	55.5	58.2	62.0	64.8	67.3
		MCDB	77.4	78.3	74.7	72.8	70.2	64.2	66.5	72.4	76.7	76.8	76.6	78.3
	5%	WB	66.8	65.8	64.4	61.5	54.8	50.4	50.0	53.4	57.1	60.7	63.6	66.1
		MCDB	76.3	77.2	72.9	71.7	68.7	62.6	65.3	70.0	75.8	76.5	75.4	77.3
	10%	WB	65.9	64.9	63.4	60.1	53.2	48.9	48.6	51.4	55.7	59.6	62.4	64.8
		MCDB	75.3	75.9	72.0	69.9	66.6	60.8	63.0	67.0	73.3	74.9	73.7	75.9
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	19.7	20.1	22.8	26.1	25.8	26.3	26.7	26.5	24.2	23.1	20.4
		MCDBR	24.2	24.6	24.5	25.6	29.7	28.6	28.8	30.0	29.9	27.4	26.7	24.9
		MCWBR	8.0	8.7	8.4	10.1	12.9	13.5	13.7	13.5	12.4	10.1	9.5	8.6
	5% WB	MCDBR	19.0	20.6	19.1	22.1	25.7	25.3	27.1	27.8	26.6	24.4	21.5	20.2
		MCWBR	7.7	8.5	7.9	10.3	12.2	12.9	13.8	13.9	11.9	10.2	9.3	8.4
(40) Clear-Sky Solar Irradiance	taub		0.371	0.376	0.361	0.319	0.283	0.267	0.268	0.320	0.362	0.391	0.346	0.355
	taud		2.374	2.355	2.377	2.466	2.542	2.555	2.529	2.373	2.265	2.223	2.403	2.408
	Ebn at Noon		308	302	296	296	294	293	297	290	291	293	314	314
	Edh at Noon		42	41	38	32	27	25	27	34	42	47	40	40
(44) All-Sky Solar Radiation	RadAvg		2156	2004	1787	1519	1380	1246	1357	1616	1949	2102	2218	2246
	RadStd		226	184	156	129	62	56	37	79	80	138	182	134

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page